

Rahul Ramachandran

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EDUCATION

University of Illinois Urbana-Champaign

PhD in Computer Science; **Advisor: Saurabh Gupta**

Aug. 2025 – Present

Indian Institute of Technology Hyderabad

Bachelor of Technology in Computer Science; **GPA: 9.89/10**

Nov. 2021 – July. 2025

PUBLICATIONS

- [1] [Rahul Ramachandran](#), Ali Garjani, Andrei Atanov, Oğuzhan Fatih Kar and Amir Zamir. "How well does GPT-4o understand vision? Solving standard computer vision tasks with multimodal foundation models", *ICLR, 2026*. Project: [Link](#)
- [2] [Rahul Ramachandran](#), Tejal Kulkarni, Charchit Sharma, Deepak Vijaykeerthy and Vineeth Balasubramanian. "On Evaluation of Vision Datasets and Models using Human Competency Frameworks", *DMLR Workshop, ICML, 2024*. Project: [Link](#)

RESEARCH & INDUSTRIAL EXPERIENCE

Gupta Lab, UIUC

Aug. 2025 – Present

Dr. Saurabh Gupta

- Developing learning-based approaches for long-horizon, contact-rich sim-to-real manipulation.
- Leveraging VR demonstrations and RL to bootstrap policies in simulation and deploy them on Franka arms.

Visual Intelligence and Learning Lab, EPFL

May. 2024 – Jul. 2024

Dr. Amir Zamir

- Assessed the performance of leading multimodal foundation models (like GPT-4o, Gemini 1.5 Pro) on standard computer vision tasks (like segmentation, depth estimation), enabling a direct comparison with specialized vision models.
- Developed several algorithms to adapt the text-based output of these models to dense visual predictions, by decomposing tasks into simpler, text-based sub-tasks and chaining the prompts.

Machine Learning & Vision Lab, IITH

Aug. 2023 – Jul. 2025

Dr. Vineeth Balasubramanian

Local Causal Discovery

- Working on a local causal discovery algorithm to identify a backdoor adjustment set for estimating average treatment effect.
- Utilizes causal identities of the treatment variable's parents and children to propose a series of tests that identify good, bad, and neutral control variables.

Item Response Theory

- Employed Item Response Theory (IRT), a human competency framework, to infer interpretable latent parameters for over a 100 vision models on ImageNet and its corrupted versions via variational inference.
- Leveraged the latent parameters to assess model calibration, select informative data subsets, and analyze models and datasets.

Research Internship at Bosch Software, Bangalore

May. 2023 - July. 2023

Dr. Sahana Prabhu

- Worked on 3D object detection in radar point clouds.
- Trained models like *Single-Stride Sparse Transformer*, *PointPillars*, *RP-Net* and worked on extending their modalities to radar.
- Improved mAP values on custom dataset through model modifications and employed DBSCAN clustering to clean the point clouds.

ACADEMIC SERVICE

DMLR Workshop Reviewer <i>International Conference on Machine Learning (ICML)</i>	2024
Undergraduate Teaching Assistant <i>Indian Institute of Technology Hyderabad</i> • Computer Architecture (CS2323) <i>Prof. Rajesh Kedia</i> • Discrete Mathematics (CS1010) <i>Prof. Rakesh Venkat</i>	2022, 2023

SCHOLASTIC ACHIEVEMENTS

NSF Graduate Research Fellowship Program (GRFP) – Prestigious fellowship awarded by the U.S. National Science Foundation to support top PhD students in STEM.	2026
Amazon AI PhD Fellowship – Awarded by Amazon and the Siebel School of Computing (UIUC) to 20 PhD students.	2025
President of India Gold Medal – Awarded for achieving the highest grade point average among all graduating B.Tech students.	2025
Summer@EPFL Fellowship – Awarded to the top 1.6% of applicants for a research fellowship at EPFL.	2024
Academic Excellence Award – Received for achieving the highest GPA in the department (awarded biennially).	2022, 2024
Joint Entrance Examination (JEE) – Secured the 229th position in JEE(Mains) and the 456th position in JEE(Advanced) out of 1,000,000 candidates.	2021
Chemistry Olympiad – State topper in Indian Olympiad Qualifier in Chemistry (IOQC) 2021.	2021
Indian National Mathematics Olympiad – Recipient of certificate of merit (awarded to the top 75 students in the country). Selection through the INMO identifies candidates for the Indian contingent in the International Mathematical Olympiad (IMO).	2020
KVPY Fellowship – Recipient of fellowship for being ranked 165th in the country; granted admission to the prestigious Indian Institute of Science (although the admission was not pursued).	2019, 2020
National Talent Search Examination – Ranked in the top 0.1% in the National Talent Search Examination, receiving a Government of India scholarship in recognition of academic achievement.	2019

TECHNICAL SKILLS

Languages: Python (advanced), C/C++ (intermediate)